School of Future Tech

Case Study Report

on

Online Auction System (C++)

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1. Introduction to the Case Study

Online auctions have become a popular method for buying and selling items through competitive bidding. In an auction system, participants place bids on an item, and the highest bidder wins the item at the end of the auction. These systems are widely used in online marketplaces and digital platforms.

This case study focuses on designing and implementing a simple Online Auction System using C++. The program allows users to enter an auction item and its base price, place bids on the item, track the highest bid, and finally declare the winner. The system demonstrates how programming logic can simulate a real-world auction process using conditional statements and loops.

The objective of this case study is to understand competitive workflows and comparison logic using C++ programming. It also helps students gain practical experience in building menu-driven applications and handling user input.

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2. Problem Statement / Case Background (Abstract)

Background

Traditional auction systems involve participants bidding competitively for an item. The bidder who offers the highest amount wins the auction. In programming, this logic can be implemented using variables, conditional comparisons, and iterative structures.

Abstract

This case study presents the design and implementation of an Online Auction System using C++. The system allows users to input an auction item and a base price. Multiple bidders can place bids, and the program continuously tracks the highest bid using comparison logic. If a new bid is greater than the current highest bid, the system updates the highest bidder and bid amount.

The application follows a menu-driven structure that provides options to place a bid, view the current highest bid, or exit the system. When the auction ends, the program declares the highest bidder as the winner. This implementation demonstrates the practical use of conditional statements, loops, and input/output operations in C++.

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3. Case Study Design

The Online Auction System is designed as a simple menu-driven console application consisting of the following stages:

1. Item Initialization

The program begins by asking the user to enter the item name and its base price.

2. Menu Display

A menu is displayed with three options:

- Place Bid
- View Highest Bid
- Exit

3. Bidding Process

Users can enter their name and bid amount. The system compares the new bid with the current highest bid.

4. Bid Validation

If the new bid is greater than the highest bid, the system updates the highest bid and bidder name. Otherwise, the bid is rejected.

5. Winner Declaration

When the user selects the exit option, the system declares the highest bidder as the winner and displays the winning bid amount.

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4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

Key Programming Methods Used

1. Conditional Logic

The program uses **if-else statements** to compare bids and determine whether the

new bid is valid.

2. Menu Control

A **switch-case statement** is used to control the program menu and execute different actions based on the user's choice.

3. Loop Structure

A **do-while loop** keeps the program running until the user decides to exit.

4. String Handling

Strings are used to store the item name and bidder names.

5. Input Validation

The program checks whether the entered bid is greater than the current highest bid.

Technology Stack Used

Programming Language:
C++

Tools and Environment:

- Dev C++ / CodeBlocks / VS Code
- Standard C++ libraries

Concepts Used:

- Variables and Data Types
- Conditional Statements
- Switch Case
- Loops
- Input/Output Operations

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5. Problem Statement / Case Study Implementation Details and Snapshots

The Online Auction System is implemented using a single C++ program file. The program contains variables for storing item details, bid values, and bidder names.

The program first asks the user to enter the auction item name and base price. After that, a menu appears where the user can choose to place a bid, view the highest bid, or exit the auction.

When placing a bid, the user enters their name and bid amount. The program compares the bid amount with the current highest bid. If the bid is valid, the highest bid and bidder name are updated.

Snapshots

```
pallavisarovar@pallavis-MacBook-Air auction sys cs % g++ online-auction-sys
pallavisarovar@pallavis-MacBook-Air auction sys cs % ./online-auction-sys
===== ONLINE AUCTION SYSTEM =====
Enter Item Name: laptop
Enter Base Price: 20000

----- MENU -----
1. Place Bid
2. View Highest Bid
3. Exit
Enter your choice: 1

Enter Bidder Name: aisha
Enter Bid Amount: 21000
Bid placed successfully!

----- MENU -----
1. Place Bid
2. View Highest Bid
3. Exit
Enter your choice: 1

Enter Bidder Name: rahul
Enter Bid Amount: 21500
Bid placed successfully!

----- MENU -----
1. Place Bid
2. View Highest Bid
3. Exit
Enter your choice: 2

Current Highest Bid: 21500
Highest Bidder: rahul

----- MENU -----
1. Place Bid
2. View Highest Bid
3. Exit
Enter your choice: 1

Enter Bidder Name: rohit
Enter Bid Amount: 23000
Bid placed successfully!
```

```
----- MENU -----
1. Place Bid
2. View Highest Bid
3. Exit
Enter your choice: 3

Auction Winner Declared!
Winning Bidder: rohit
Winning Bid: 23000
Thank you for using the Auction System.
pallavisarovar@pallavis-MacBook-Air auction sys cs %
```

These screenshots demonstrate the working of the auction system.

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6. Problem Statement / Case Study Results and Conclusion

Results

The Online Auction System successfully performs the following tasks:

- Accepts item details and base price
- Allows users to place bids
- Validates bids using comparison logic
- Tracks the highest bidder
- Declares the winner when the auction ends

The program works as a menu-driven application and simulates a simple competitive auction environment.

Conclusion

This case study demonstrates how a real-world auction process can be implemented using C++ programming. The system uses conditional statements, loops, and menu-driven logic to manage bidding operations.

Through this project, the concepts of comparison logic, user input handling, and structured programming were applied effectively. The Online Auction System provides a clear example of how programming can be used to simulate practical real-world systems.

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7. References

- C++ Programming Language Documentation
- Standard C++ Library Reference
- Online tutorials and programming resources related to menu-driven C++ programs
- Classroom notes and learning materials for C++ programming